Problem 11 | Problems T4_PhaseDiagrams I.pdf #11 | Diagram construction

Drawing a Diagram with a Stoichiometric Compound

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 4

11. Draw the hypothetic phase diagram of A and B from the following data. a) Melting temperatures: A = 850 °C; B = 350 °C. b) A and B form an stoichiometric compound AB2 that melts at 1000 °C; neither A nor B are soluble in the compound. c) There is an invariant eutectic point (70% at. of A, 700 °C). d) There is an invariant peritectic point (80% at. of B, 500 °C). e) The composition of the liquid at 500 °C is 95% at. of B. f) The solubility of B in A is 10% at. at 700 °C and 2% at. at 25 °C. g) The solubility of A in B is 25 % at. at 25 °C. Connect the points with straight lines. Indicate all the phase areas and points in the diagram, and write the reactions of the three phases.

Official Solution: not given in the sheet

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. Compositions in **atomic percent of B** (at.% B) on the horizontal axis; straight lines between the given points, as the statement asks.
2. **AB2** contains two B atoms per A atom: $x_B = 2/3 = 66.7$ at.% B. It is stoichiometric and dissolves neither A nor B, so it is a **vertical line** [Session 8 Slide 15]; it melts at 1000 °C as a single phase, so it melts **congruently** (a maximum of the liquidus) [Session 8 Slide 17].
3. The terminal solid solutions are α (B dissolved in A) and β (A dissolved in B).
4. "80 % at. of B" for the peritectic point is the composition of the solid β formed there; the liquid at 500 °C has 95 at.% B (point e).

Points

Point	at.% B	T (°C)
Melting point of A	0	850
Melting point of B	100	350
AB_2 (congruent melting)	66.7	1000
Eutectic E (liquid)	30	700
α at E (max. solubility of B in A)	10	700
α at 25 °C	2	25
Peritectic P (β formed)	80	500

Point	at.% B	T (°C)
Liquid at 500 °C	95	500
β at 25 °C (25 at.% A)	75	25

PHASE 2 · FRAMES & STRATEGY

Frame: composition (at.% B) against temperature (°C). The diagram is built from straight segments between the points; no change of basis is involved.

How the points connect.

- Left part (A to AB_2): a **eutectic** system between α and AB_2 [Session 7 Slide 26]: liquidus from A (0, 850) to E (30, 700) and from E up to AB_2 (66.7, 1000); α solidus from (0, 850) to (10, 700); eutectic line from α (10) to AB_2 (66.7) at 700 °C; α solvus from (10, 700) to (2, 25).
- Right part (AB_2 to B): a **peritectic** system [Session 8 Slide 8]: liquidus from AB_2 (66.7, 1000) down to (95, 500) and then to B (100, 350); peritectic line from AB_2 (66.7) to L (95) at 500 °C, with β (80) on it; β solidus from (80, 500) to B (100, 350); β solvus from (80, 500) to (75, 25).

PHASE 3 · STEP-BY-STEP DERIVATION

The diagram

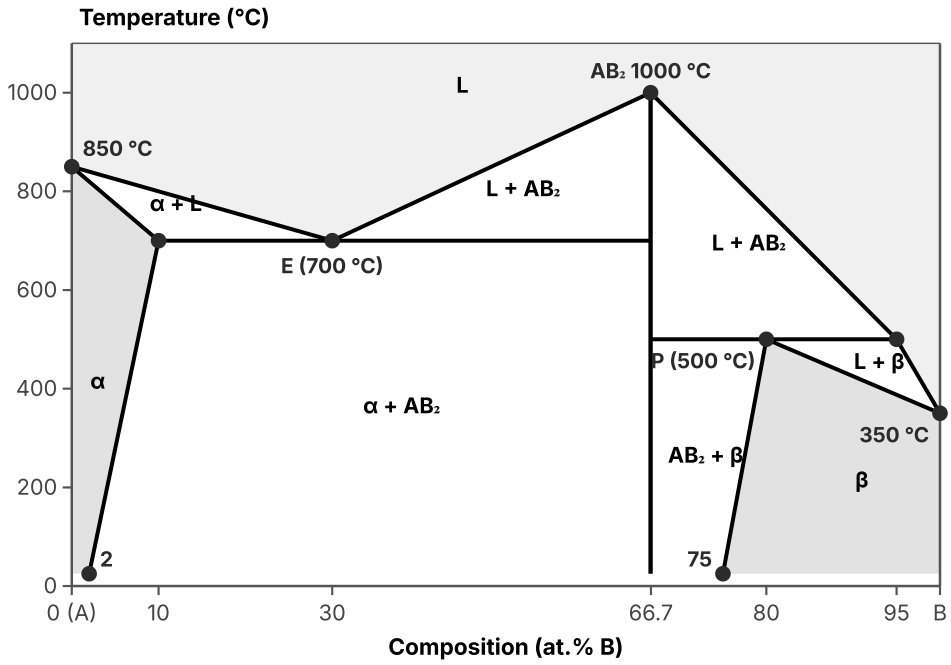


Figure 18. A-B with straight lines: A melts at 850 °C, B at 350 °C, the stoichiometric compound AB_2 (66.7 at.% B) melts congruently at 1000 °C; eutectic E at 700 °C and 30 at.% B; peritectic P at 500 °C (β of 80, liquid of 95 at.% B); solubility of B in α 10 at.% at 700 °C and 2 at.% at 25 °C; solubility of A in β 25 at.% at 25 °C.

Phase areas

Region	Bounded by
L	above the liquidus
α + L	0-30 at.% B, between 850 and 700 °C
α	left of the α solidus and solvus
α + AB_2	10-66.7 at.% B, below 700 °C
L + AB_2 (left)	30-66.7 at.% B, between 700 °C and the liquidus
L + AB_2 (right)	66.7-95 at.% B, between 500 °C and the liquidus
AB_2 + β	66.7 at.% B to the β solvus, below 500 °C
L + β	between the β solidus (80 → 100) and the liquidus (95 → 100), from 500 to 350 °C
β	right of the β solidus and solvus

Reactions of three phases (on cooling)

- **Eutectic, 700 °C:** $L\,(30\ \text{at.\%\ B}) \rightarrow \alpha\,(10\ \text{at.\%\ B}) + \text{AB}_2\,(66.7\ \text{at.\%\ B})$
- **Peritectic, 500 °C:** $\text{AB}_2\,(66.7\ \text{at.\%\ B}) + L\,(95\ \text{at.\%\ B}) \rightarrow \beta\,(80\ \text{at.\%\ B})$

AB2 melting at 1000 °C involves only two phases (AB2 and a liquid of the same composition), so it is not a three-phase reaction.

PHASE 4 · CHECKS & INTERPRETATION

Checks

- Each three-phase line has the middle phase in the right place: at 700 °C the liquid (30) lies **between** the two solids (10 and 66.7), the shape of a eutectic; at 500 °C the new solid β (80) lies between AB2 (66.7) and L (95), with the liquid at one end, the shape of a peritectic [Session 8 Slide 6].
- Phase rule: on each horizontal line three phases coexist, $F = 2 - 3 + 1 = 0$; in each two-phase area $F = 1$ [Session 7 Slide 13].
- **Limiting case** (compositions in the units of the statement): at the composition of the stoichiometric compound the alloy is a single phase, so every lever-rule fraction there is 0 or 1; this is why the compound is drawn as a vertical line.

Interpretation The compound AB2 splits the diagram into two simpler diagrams, A-AB2 (eutectic) and AB2-B (peritectic). Since B melts at only 350 °C, the liquid persists to low temperatures on the B side, and β forms from AB2 and the B-rich liquid at 500 °C.

Check against the official solution

The sheet gives no solution for this problem and the official solutions PDF has no page for it, so there is nothing to compare with.